

Proactive Care Dashboards: Recommendations for Ageing-in-Place

Marina Vicini
Aston University
Birmingham, United Kingdom
m.vicini@aston.ac.uk

Ahmad Beltagui
Advanced Services Group
Aston University
Birmingham, United Kingdom
a.beltagui@aston.ac.uk

Abstract

As the global population ages, Proactive care can be a means to achieve societal benefit through Ageing-in-place, yet technology is currently only used in a traditional, reactive form. It leverages ambient sensing to detect behavioural anomalies for early health intervention. However, uptake remains low due to significant adoption barriers due to complex human and organisational networks. This research employs a qualitative, mixed-method approach, including a 24-month longitudinal observation within a technology manufacturer and focus groups with 20 professional service providers from four UK-based organisations, to identify technical, usability, and organisational recommendations for proactive care dashboards. The analysis reveals critical frictions across user, service, and value-network levels, such as environmental noise and platform fatigue. We propose that effective dashboards must evolve from passive displays into active boundary objects. Key recommendations include implementing robust, incremental learning algorithms, supporting human-in-the-loop annotations, and creating integrated action pathways to bridge organisational silos.

CCS Concepts

• **Applied computing** → Life and medical sciences; Health care information systems; • **Social and professional topics** → User characteristics; Age; Seniors; • **Human-centered computing** → Ubiquitous and mobile computing; Ubiquitous and mobile computing theory, concepts and paradigms; Ambient intelligence; Human computer interaction (HCI); Interactive systems and tools.

Keywords

Proactive care, Ageing-in-place, Dashboards, Socio-technical systems

ACM Reference Format:

Marina Vicini and Ahmad Beltagui. 2026. Proactive Care Dashboards: Recommendations for Ageing-in-Place. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3798908>



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ACM ISBN 979-8-4007-2281-3/2026/04

<https://doi.org/10.1145/3772363.3798908>

1 Introduction

Ageing-in-place, the ability for individuals to live in their own homes and communities safely and independently, is becoming a global necessity. By 2050, the population aged over 65 is projected to reach 1.6 billion, accounting for 16% (1 in 6) of the world's population [1]. Beyond managing healthcare costs by avoiding hospital admissions, it fundamentally supports the autonomy, self-determination, and dignity of the individual [2], [3]. Technology is a vital enabler; however, traditional care has been reactive, relying on alarms triggered only after an incident occurs, and Alarm Receiving Centres (ARC) determining escalation. The UK current transition from analogue to digital solutions enables proactive care [4], by using ambient sensing data to identify subtle behavioural shifts, such as changes in sleep or mobility, to predict declining health before a crisis occurs [5]. Despite this potential, uptake remains limited due to the complexity of the human and organisational networks involved in cooperative care. Care is recognised as deeply cooperative work involving a heterogeneous network of relatives, clinicians, and equipment installers, which varies in structure and communication [6]. Successful technology-enabled care must be *co-produced*, ensuring services align with the messy reality of domestic life [7]. This necessitates significant articulation work: the ongoing labour of adjusting physical devices to unique environments and the 'invisible effort' required to coordinate distributed tasks and handle contingencies [8], [9].

The primary interface for this coordination is the proactive care dashboard, which serves as a tool for knowledge exchange on the individual's status. These platforms integrate data from ambient sensors (e.g., motion detectors, door alarms, smart meters [10], [11]), environmental sensors (e.g., heat or humidity [12]), and/or physiological markers (e.g., blood pressure monitoring [13]) with anomaly-detection algorithms. Current dashboards frequently suffer from a lack of usability or explainability, leaving a significant user experience research gap [7], due to tensions between developers and the users who must interpret them [14] and limits to interorganisational collaboration due to limits of data exchange.

Our work addresses these gaps by adopting a customer-centred approach to provide recommendations that improve algorithm and service delivery. Our contributions are as follows:

- We present a rich empirical account of the technical and organisational challenges inherent in complex, data-driven proactive care services.
- We identify critical design tensions at the user-interaction, service-interaction, and network-interaction levels, from which we identify technical and usability recommendations for proactive care.

Our distinct novelty lies in utilising a professional-stakeholder lens to surface specific operational frictions, thereby reframing proactive dashboard design not merely as a user interface challenge, but as essential coordination infrastructure. Our evidence is UK specific, but the need for proactive care for Ageing-in-place is reported and investigated in other countries as well [15], [16], expanding the potential impact of this work outside of the UK.

2 Background

A proactive care dashboard, variously referred to in literature as a connected care platform [7], Activities of Daily Living (ADL) platform, or Decision Support System (DSS) [17], is defined as an interface that provides graphical representations of current status, historical trends, and key performance indicators to support decision-making [18]. Beyond its technical function, the dashboard acts as a cognitive assistant and a boundary object [19]. It serves as a mediator that allows diverse actors to negotiate and integrate knowledge across social, symbolic, and material boundaries within a care network.

2.1 User target groups

Literature distinguishes between several primary user groups, each with distinct informational needs:

- End users (older adults): The platform shares personal data back to the individual to support self-management of wellbeing and safety [20], [21], [22]. Visualisations provide feedback that allows consumers to engage with their own data [23], though care must be taken to avoid misinterpretations that could cause unnecessary worry [20].
- Family carers: They use the dashboard for reassurance on the health of the relatives. They typically check the dashboard more frequently and require more granular, detailed information than care providers [16].
- Care providers: Responsible for a larger number of individuals, they prioritise identifying ‘at-risk’ users. They require data to be abstracted into actionable information (e.g., colour-coded risk level) to manage care plans [24].
- Organizational managers: These stakeholders focus on resource allocation and service overviews rather than the specific care needs of an individual [24].

Literature focuses on technology developers and end-users (older adults), whereas the service providers who specify, procure, install and operate tech are underrepresented [7], [20].

2.2 Characteristics of the proactive dashboard

A core challenge identified in the literature is the ‘hidden work’ generated by these technologies for service providers, as new roles and responsibilities are created for social care workers. This effort involves sense-making: the cognitive effort required to determine if a data anomaly is clinically significant, integrating that data with existing knowledge of the individual, and coordinating a response across the care network [17].

The primary vehicle for this sense-making is the visualisation of longitudinal data. Designers can choose between varied visualisation strategies (e.g., activity heatmaps, traffic light systems, or radial plots) depending on the level of abstraction or temporal

requirements [22], [23], [24], [25]. We can distinguish between wellbeing dashboard, focused on ADL [16], and clinical dashboard focused on physiological markers, medication discrepancies, and fall risk scores [24], [26].

The lack of real-world datasets with ground truth of an individual’s health status is a consistently reported challenge [27], increasing the difficulty of evaluating the robustness of algorithms within complex environments. This gap ultimately hinders the assessment of the effectiveness of the platform by service providers to support vulnerable individuals.

2.3 Requirements for effective deployment

Based on existing literature and operational frictions, several critical technical requirements emerge for proactive care solutions. First, the system must be *hierarchical*, supporting the different target roles by displaying information accordingly, and allowing for ‘drill-through’ capabilities that allow a user to move from a population-level macro view to specific event-level details [24]. Second, *explainability* is paramount; dashboards should provide natural language explanations and transparent decision-making logic alongside graphical data to help users understand why an anomaly was flagged [16]. The decision-making must be supported by accuracy of the data.

Finally, the platform must facilitate *collaborative workflows*. This includes the ability for various actors to confirm alerts, contextualise data, or keep track of actions taken [17]. In addition, effective platforms allow for data sharing, by attempting to bridge the data silos by linking administrative, clinical, and sensor-based datasets into a unified interface [24], [28]. While personalisation is necessary to align with the ‘messy reality’ of domestic life, designers must balance this against the need for standardised solutions that can be scaled across large-scale social care deployments [17], [29]

3 Methodology

To investigate the design challenges of Proactive Care, this research used a qualitative, mixed-method approach rooted in Engaged Scholarship [30]. This methodology involves co-producing knowledge with practitioners from within the problem domain to better understand the commercial and operational realities of technological deployment. The study received ethics approval from Aston University (REC ID: BSS21289), and informed consent was obtained from all participants. The research was conducted in two primary phases: longitudinal participant observation and external focus groups.

3.1 Longitudinal participant observation

The first author was embedded within the product development team of a European Independent and Assisted Living Technology manufacturer for 24 months (2024–2025). This allowed for real-time observation of decision-making processes regarding the technical development of behaviour detection algorithms and the design of monitoring solutions. This longitudinal participant observation was primarily used for contextualisation and triangulation of the systemic challenges. Rather than generating specific themes, it allowed us to validate the focus group findings against the practical,

Table 1: Overview of the 20 study participants from UK-based organisations

Role category	Count	ID ranges	Example
Strategic	8	P1-P3, P5, P8, P14-P15, P20	Telecare service managers, digital leads
Technical	7	P9-13, P16-17	Equipment Installers
Managerial/Ops	4	P6-7, P18-19	ARC Managers, Coordinators
Clinical	1	P4	Occupational therapist

commercial, and technical realities of bringing proactive solutions to market.

3.2 External focus groups

To capture the requirements of the broader social care ecosystem, we conducted four semi-structured focused groups with UK organisations responsible for commissioning assisted living technologies and managing. The study utilised an internal customer segmentation workshop as a purposive sampling mechanism. Members of the manufacturer’s UK sales team, acting as key account managers, identified promising customer organisations based on their digital maturity and strategic focus on innovative technology delivery. This ensured that the focus groups involved individuals capable of reflecting deeply on the transition to proactive care, as they would be familiar with the technology and possess the infrastructure and skills to adopt it. However, this purposive sampling strategy introduces a bias towards technology-ready organisations. This potentially excludes those that struggled most with, or rejected, proactive care adoption, which could offer alternative insights into failure modes.

Four semi-structured focus groups were conducted with 20 participants from four distinct UK-based organisations (we refer to them by pseudonyms P1-P20), as can be observed in Table 1. These organisations included two local authorities with integrated health and social care boards, one local authority with separate departments, and a third-sector charity. The sessions totalled 721 minutes of audio recording (averaging 180 minutes per session). To avoid a narrow focus on specific products, the discussions were framed around five broad questions regarding external threats, strategic targets, interaction processes with older adults, service delivery restrictions, and primary organisational concerns. The study intentionally focused on these professional stakeholders rather than end users. The aim is to better understand the needs of those providing care, and their challenges when interacting with the user. In addition, these individuals act as the primary decision-makers for technology adoption and possess aggregated insights into the organisational challenges.

3.3 Data Analysis

All focus groups were audio-recorded and transcribed. We applied a thematic analysis approach [31] to identify patterns across this heterogeneous data. The analysis was an inductive, iterative and collaborative process, involving several key phases. Initially, an inductive thematic analysis was conducted on the dataset to identify broad systemic challenges, resulting in three themes: the user, service, and value-network levels. Building on these findings,

we performed a second, targeted iteration to address the specific research question: “*What are the technical and usability requirements of a proactive care dashboard?*”. This study does not provide a comprehensive requirement list of a proactive care solution, but significant recommendations. The three themes extracted during the first iteration on the data remained valid in answering this more specific research question. Throughout the process, the 24-month longitudinal observation served to contextualise findings, ensuring focus group themes aligned with the manufacturer’s real-world technical decision-making. In addition, we specifically analysed how the dashboard functions as a boundary object by looking for properties such as interpretive flexibility and knowledge negotiation across care networks and identifying where coordination breaks down.

4 Results

Through the analysis, the three levels of interactions previously identified, i.e. the user level, the service level and the value network one, still emerged as the main themes of the data, that were explored by interrogating the new research question. With user level, we refer to the recipient of the service, i.e. an older adult living independently at home.

4.1 User-level

At the user level, participants reported on their service delivery experience with older adults. Participants across different organisations reported that most proactive solutions they encountered are unprepared for pets in the apartment, triggering false alerts, or were unsuitable for multi-occupancy environment, limiting their deployment. Beyond environmental factors, user behaviour significantly impacts data continuity. For instance, P1 noted that older adults frequently switch off wall plugs in the evening to prevent electric fires, inadvertently disabling monitoring and triggering unnecessary alerts to the ARC. Such actions may stem from mental models or cognitive issues, where users forget the technology’s purpose. To mitigate these issues, P20 suggested implementing ‘dynamic consent’ tools that allow users to pause monitoring manually. Strategic timing is also critical; installers P16 and P17 reflected that technology must be introduced early, well before a health crisis, to allow for learning and lifestyle integration. A modular approach is favourable, where telecare devices are incrementally included based on the person needs, to avoid overprescription. Installers use ‘hooks’ (e.g., jewellery) to integrate telecare into a user’s lifestyle and ensure compliance despite the associated stigma.

Table 2: Overview of interaction levels, key challenges, and recommendations.

Level	ID ranges	Example
User	Environmental noise; connectivity/consent gaps.	Robust, incremental learning algorithms; dynamic consent tools.
Service	High cognitive load; ‘hidden work’ in sense-making; trust erosion.	Human-in-the-loop annotations; display uncertainty.
Network	Platform fatigue; lack of clear action pathways.	Integrated boundary objects; automated workflow triggers.

4.2 Service-level

At the service level, the primary challenge involves managing the ‘hidden work’ created for professional carers and ARC operators. Data accuracy is paramount because fundamental life decisions for vulnerable individuals are based on these outputs. A telecare manager (P20) emphasised that erroneous information from a single faulty sensor undermines confidence in the entire system. Team managers (P18, P19) reported that ARC operators struggle to respond to alarms as they often lack visual and contextual information, leading to discrepancies between sensor data and a user’s verbal report. Without sufficient contextual data to support interpretive flexibility, operators and end-users cannot effectively negotiate the clinical reality of the alert. Furthermore, there is a strong demand for ‘human-in-the-loop’ functionality, allowing staff to track, flag, and annotate alerts with clinical context. Participants in all organisations reiterated that simplicity in design is non-negotiable; limited time and limited IT literacy requires actionable data and digestible report, to not increase the cognitive load on already-strained staff, leading to platform abandonment if the workload becomes unsustainable.

4.3 Value network level

The value network level focuses on the systemic integration of these tools within the broader care ecosystem. Participants expressed significant ‘platform fatigue,’ with P2 stating they do not want to monitor multiple standalone dashboards and prefer all data to sit within a single service user file. Currently, data sources often fail to integrate or do so only at prohibitive costs. P20 envision the dashboard as a shared boundary object, a central, standardised space where social care teams, informal carers, and clinicians access the same information, but apply their distinct expertise, ensuring that hospital admissions or care plan changes are updated in one place rather than across multiple silos. Relatives use the dashboard to check-in on their loved ones, and service providers to understand users most at risk.

However, gathering data is only valuable if a clear ‘action pathway’ exists. Participants raised concerns on having the alerts raised without pathways to act on it or without resources could create a heavy burden, such that P18 and P19 shared that these alerts are better suited for family or clinicians. This represents a failure in cross-boundary negotiation; without predefined action pathways, the dashboard surfaces information but fails to coordinate the distributed tasks necessary to deliver care. New roles were reported in one organisation, with a new welfare team appointed to overview the longitudinal data to best address the care plan.

5 Discussion

Proactive care success depends on evolving dashboards into active boundary objects that facilitate knowledge negotiation across care networks [19]. To address operational frictions, we propose several recommendations (Table 2).

From the end-user interaction with devices, algorithms must be robust to environmental noise caused by pets or multi-occupancy. Furthermore, the system must handle missing data resulting from intermittent connectivity or the withdrawal of consent. Rather than treating these as system failures, technical requirements must include data imputation methods that respect dynamic consent, thereby upholding the user’s need for autonomy and dignity [2]. The requirement for incremental learning, driven by modularly introducing telecare, means that systems should adapt without re-training from scratch. As identified in previous studies, technology is socially leveraged [32]; it does not consist of independent incidents but requires continuity [14] to be socially dependable [33], meaning it must respect and fit within a user’s social environment.

The dashboard must facilitate a ‘human-in-the-loop’ approach, allowing family or professional carers to flag or confirm alerts, and these flags can serve as new data points for *reinforcement* learning algorithms. The dashboard is aimed to augment, not automate decisions, and human professional judgment remains the critical link [17]. Given that erroneous information can undermine trust, systems must ensure high data accuracy. Anomalous data points due to device positioning, low battery or more, must be flagged as uncertain, to increase reliability. Additionally, any time delays in near-real-time information must be clearly displayed. If a delay is not communicated, ARC operators may treat the data as real-time; any discrepancy between the dashboard and what the user says during a call can undermine trust. Call operators perform significant emotional labour and possess vital local knowledge [34]; technology is aimed to enhance knowledge exchange to improve care, rather than increasing hidden work due to sense-making [17].

A core challenge is that these technologies often generate hidden work for service providers, creating new roles and responsibilities for social care workers. This involves sense-making: determining if a data anomaly is clinically significant, integrating it with existing knowledge of the individual, and coordinating a response. The dashboard must act as a boundary object that bridges organisational silos by serving as a central data-sharing point for clinical, administrative, and wellbeing data [24], [28]. This is challenging due to conflicting objectives between organisations. The current lack of pathway highlights that the dashboard does not live in a

vacuum, and it must be co-produced by the care network, overcoming competing stakeholder interests [35]; otherwise even the most sophisticated data remains a burden rather than a benefit.

6 Conclusion

This research demonstrates that the transition to proactive care is a complex socio-technical effort. Analysis of operational frictions across user, service, and value-network levels reveals that effective dashboards must evolve into active boundary objects. These tools must be robust enough to handle the ‘messy’ realities of domestic life, such as environmental noise and dynamic consent, to support and augment the professional judgment of care providers. Ultimately, the value of proactive data is only realised when it is integrated into clear action pathways that bridge organisational silos. We intend to translate these recommendations into a functional prototype to serve as a foundation for iterative co-design exercises with practitioners and end users. By acting as a central hub for cross-organisational collaboration, the dashboard facilitates cooperative work, enabling diverse actors to leverage their specific expertise and responsibilities for mutual benefit.

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